

AP CSP | Lesson 18: Traversing Lists

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5: select exactly TWO. FOR EACH visits each element value in order. Keep the source list unchanged; append matches to a new list. Code braces go on separate lines.

1. For `a ← [9, 4, 4]`, what is `a[2]`? [1 mark]

- A. 9 B. 4 C. 2 D. 3

Answer: _____

2. FOR EACH item IN `[10, 30, 20]` assigns which values to item in order? [1 mark]

- A. 1, 2, 3 B. 3, 2, 1 C. 10, 30, 20 D. 10, 20, 30

Answer: _____

3. Where should `total ← 0` be placed to sum an entire list? [1 mark]

- A. Inside IF after each match B. After displaying the total
C. At the end of every visit D. Before the traversal

Answer: _____

4. How many loop-body executions occur for FOR EACH item IN `[6, 6, 6]`? [1 mark]

- A. 3 B. 1 C. 6 D. 0

Answer: _____

5. Which TWO statements match this lesson's filter design? Select exactly TWO answers. [1 mark]

- A. Remove nonmatches from the source during traversal. B. Initialize a separate result as `[]`.
C. Append each matching occurrence to that result. D. Sort the source automatically before filtering.

Answer: _____

6. `total` starts at 0, then each value in `[]` is added to it. What is the final total? [1 mark]

- A. An invalid index is always accessed B. 1
C. The last element D. 0

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

Start total=0. Add each value in [5, 10, 5] during FOR EACH. Give (a) total after visit 1, (b) after visit 2, (c) after visit 3, and (d) the number of visits.

8. [4 marks]

Start selected=[]. For each value in [12, 30, 30, 45], append it when value ≥ 30 . Give selected after each of the four visits.

9. [4 marks]

Write code that sets total to 0, visits each minutes value in sessions, adds it to total, and displays the final total after traversal. Assume sessions is a numeric list.

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Written responses | Questions 10-12

10. [4 marks]

Write code that creates `selected=[]` and appends each value from `sessions` that is at least `goal`. Preserve `sessions` and its order. `sessions` is numeric and `goal` is a number.

11. [4 marks]

A filter keeps values ≥ 30 . Give the resulting list for (a) [29], (b) [30], (c) [31], and (d) [].

12. [4 marks]

For each value in [4, 7, 2], a program first sets `total=0` and then adds the value to `total`. (a) Give its final total. (b) State the fix. (c) Give the corrected total. (d) Explain why changing the source during a separate filtering traversal is avoided here.
